



HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



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FUNDAMENTALS OF OPTIMIZATION

MODELLING

ONE LOVE. ONE FUTURE.

CONTENT

- Modelling
- Constraint linearization
- Examples

- Modelling consists of specifying
 - Decision variables
 - Constraints
 - Objective functions
- A problem can be modelled in different ways (how to define variables)
- Take into account the modelling languages of software tools
 - Constraint Programming solvers: constraints can be stated in flexible ways

CONSTRAINT LINEARIZATION

- Motivation
 - Linear Programming solvers are very efficient
- Examples
 - How to model $X = \min\{x_1, x_2\}$?

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Solution: define an auxiliary binary variable y , use big constant M

- $x_1 \geq X$
- $x_2 \geq X$
- $X \geq x_1 - M(1-y)$
- $X \geq x_2 - My$

CONSTRAINT LINEARIZATION

- Examples
 - How to model $(x = 1) \Rightarrow (z \geq y)$ where x is binary variable, y and z are real variables ?

CONSTRAINT LINEARIZATION

- Examples
 - How to model $(x = 1) \Rightarrow (z \geq y)$ where x is binary variable, y and z are real variables ?

Solution: use big constant M

- $M(x-1) + y \leq z$

CONSTRAINT LINEARIZATION

- Example
 - $(x > 0) \Rightarrow (z \geq y)$ in which x, y and z are real variables (and $x \geq 0$) ?

CONSTRAINT LINEARIZATION

- Example
 - $(x > 0) \Rightarrow (z \geq y)$ in which x, y and z are real variables (and $x \geq 0$) ?

Solution:

- Let M be a very big constant,
- Introduce a binary variable $t \in \{0,1\}$:
 - $t = 1$ indicates that $x > 0$, and $t = 0$ indicates that $x = 0$
- Equivalent linear constraints
 - $x \leq M.t$
 - $z + (1-t)M \geq y$

- **Balanced Course Assignment Problem**

- At the beginning of the semester, the head of a computer science department D have to assign courses to teachers in a balanced way. The department D has m teachers $T=\{0, 2, \dots, m-1\}$ and n courses $C=\{0, 2, \dots, n-1\}$.
 - Each teacher $t \in T$ has a preference list which is a list of courses he/she can teach depending on his/her specialization. The preference information is represented by a 0-1 matrix $A_{m \times n}$ in which $A(t,c) = 1$ indicates that teacher t can teach the course c and $A(t,c) = 0$, otherwise
 - We know a set B of pairs of conflicting two courses that cannot be assigned to the same teacher as these courses have been already scheduled in the same slot of the timetable.
 - The load of a teacher is the number of courses assigned to her/him. How to assign n courses to m teacher such that each course assigned to a teacher is in his/her preference list, no two conflicting courses are assigned to the same teacher, and the maximal load among teachers is minimal.

EXAMPLES

- Balanced Course Assignment Problem

Course	0	1	2	3	4	5	6	7	8	9	10	11	12
credits	3	3	4	3	4	3	3	3	4	3	3	4	4

Teachers	Preference Courses
0	0, 2, 3, 4, 8, 10
1	0, 1, 3, 5, 6, 7, 8
2	1, 2, 3, 7, 9, 11, 12

Conflict courses

0	2
0	4
0	8
1	4
1	10
3	7
3	9
5	11
5	12
6	8
6	12

EXAMPLES

- Balanced Course Assignment Problem

Course	0	1	2	3	4	5	6	7	8	9	10	11	12
credits	3	3	4	3	4	3	3	3	4	3	3	4	4

Teachers	Preference Courses
0	0, 2, 3, 4, 8, 10
1	0, 1, 3, 5, 6, 7, 8
2	1, 2, 3, 7, 9, 11, 12

Teacher	Assigned courses	Load
0	2, 4, 8, 10	15
1	0, 1, 3, 5, 6	15
2	7, 9, 11, 12	14

Conflict courses

0	2
0	4
0	8
1	4
1	10
3	7
3	9
5	11
5	12
6	8
6	12

- **Balanced Course Assignment Problem: A Constraint Programming model**
 - Decision variables
 - $X(i)$: teacher assigned to course i , $\forall i \in C$, domain $D(X(i)) = \{t \in T \mid A(t,i) = 1\}$
 - $Y(i)$: load of teacher i , domain $D(Y(i)) = \{0,1,...,n-1\}$
 - Z : maximum load among teachers
 - Constraints
 - $X(i) \neq X(j), \forall (i,j) \in B$
 - $Y(i) = \sum_{j \in C} (X(j) = i), \forall i \in T$
 - $Z \geq Y(i), \forall i \in T$
 - Objective function to be minimized: Z

- **Balanced Course Assignment Problem: An Integer Linear Programming model**
 - Decision variables
 - $X(i,j) = 1$: teacher i is assigned to course j , and $X(i,j) = 0$, otherwise, $\forall i \in T, j \in C$, domain $D(X(i,j)) = \{0,1\}$
 - $Y(i)$: load of teacher i , domain $D(Y(i)) = \{0,1,...,n\}$
 - Z : maximum load among teachers
 - Constraints
 - $\sum_{i \in T} X(i,j) = 1, \forall j \in C$
 - $X(t,i) + X(t,j) \leq 1, \forall (i,j) \in B, t \in T$
 - $Y(i) = \sum_{j \in C} X(i,j), \forall i \in T$
 - $Z \geq Y(i), \forall i \in T$
 - Objective function to be minimized: Z

- **Travelling Salesman Problem (TSP)**

- A salesman departs from point 1, visiting $N-1$ points 2, ..., N and comes back to the point 1. The travelling distance from point i and point j is $d(i,j)$, $i,j = 1,..., N$. Compute the route of minimal total travelling distance

- **Travelling Salesman Problem (TSP): An Integer Linear Programming model**

- Decision variables

- Binary variable $X(i,j) = 1$ if the route traverses from point i to point j , and $X(i,j) = 0$, otherwise.

- Constraints

- $\sum_{j=1}^N X(i,j) = \sum_{j=1}^N X(j,i) = 1, \forall i \in \{1,2,\dots,N\}$
- $\sum_{(i,j) \in S} X(i,j) \leq |S| - 1, \forall S \subseteq \{1,2,\dots,N\} \text{ and } |S| < N$

- Objective function to be minimized

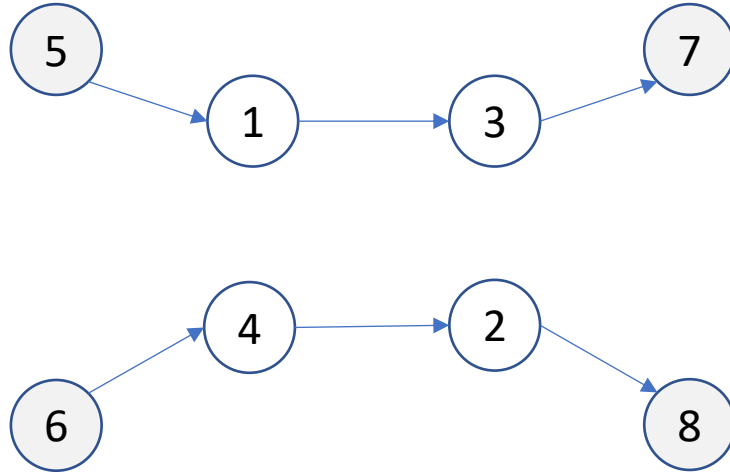
$$f(X) = \sum_{j=1}^N \sum_{i=1}^N d(i,j)X(i,j)$$

- **Capacitated Vehicle Routing Problem**

- A fleet of K trucks $1, 2, \dots, K$ must be scheduled to visit N customers $1, 2, \dots, N$ for collecting items
 - Customer i located at point i and requests to be collected $r(i)$ items, $i = 1, 2, \dots, N$
 - Truck k ($k = 1, \dots, K$)
 - Departs from point $N+k$ and terminates at point $N + K + k$ ($N+k$ and $N+K+k$ might refer to the central depot)
 - has capacity $c(k)$ which is the maximum number of items it can carry at a time
 - Travel distance from point i to point j is $d(i,j)$, $i,j = 1, \dots, N + 2K$
- Compute the delivery solution such that the total travelling distance is minimal
 - Satisfy capacity constraints
 - Each customer is visited exactly once by exactly one truck

EXAMPLES

- Capacitated Vehicle Routing Problem



1	0	2	3	4	3	3	3	3
2	4	0	2	6	1	1	1	1
3	2	4	0	2	1	1	1	1
4	5	7	7	0	4	4	4	4
5	3	1	5	7	0	0	0	0
6	3	1	5	7	0	0	0	0
7	3	1	5	7	0	0	0	0
8	3	1	5	7	0	0	0	0

	1	2
c	10	10

	1	2	3	4
r	4	3	6	5

- **Capacitated Vehicle Routing Problem**

- Notations

- $B = \{1, \dots, N+2K\}$
- $F_1 = \{(i, k+N) \mid i \in B, k \in \{1, \dots, K\}\}$
- $F_2 = \{(k+K+N, i) \mid i \in B, k \in \{1, \dots, K\}\}$
- $F_3 = \{(i, i) \mid i \in B\}$
- $A = B^2 \setminus F_1 \setminus F_2 \setminus F_3$
- $A^+(i) = \{j \mid (i, j) \in A\}, A^-(i) = \{j \mid (j, i) \in A\}$

- Decision variables

- $X(k, i, j) = 1$ if truck k travel from point i to point j , $\forall k = 1, \dots, K, (i, j) \in A$
- $Y(k, i)$: number of items on truck k after leaving point i , $\forall k = 1, \dots, K, \forall i = 1, \dots, N+2K$
- $Z(i)$: index of truck visiting point i , $\forall i = 1, 2, \dots, N+2K$

- **Capacitated Vehicle Routing Problem**

- Constraints

- $\sum_{k=1}^K \sum_{j \in A^+(i)} X(k, i, j) = \sum_{k=1}^K \sum_{j \in A^-(i)} X(k, j, i) = 1, \forall i = 1, \dots, N$
- $\sum_{j \in A^+(i)} X(k, i, j) = \sum_{j \in A^-(i)} X(k, j, i), \forall i = 1, \dots, N, k = 1, \dots, K$
- $\sum_{j=1}^N X(k, k + N, j) = \sum_{j=1}^N X(k, j, k + K + N) = 1, \forall k = 1, \dots, K$
- $M(1 - X(k, i, j)) + Z(i) \geq Z(j), \forall (i, j) \in A, \forall k = 1, \dots, K$
- $M(1 - X(k, i, j)) + Z(j) \geq Z(i), \forall (i, j) \in A, \forall k = 1, \dots, K$
- $M(1 - X(k, i, j)) + Y(k, j) \geq Y(k, i) + r(j), \forall (i, j) \in A, \forall k = 1, \dots, K$
- $M(1 - X(k, i, j)) + Y(k, i) + r(j) \geq Y(k, j), \forall (i, j) \in A, \forall k = 1, \dots, K$
- $Y(k, k + K + N) \leq c(k), \forall k = 1, \dots, K$
- $Y(k, k + N) = 0, \forall k = 1, \dots, K$
- $Z(k + N) = Z(k + K + N) = k, \forall k = 1, \dots, K$

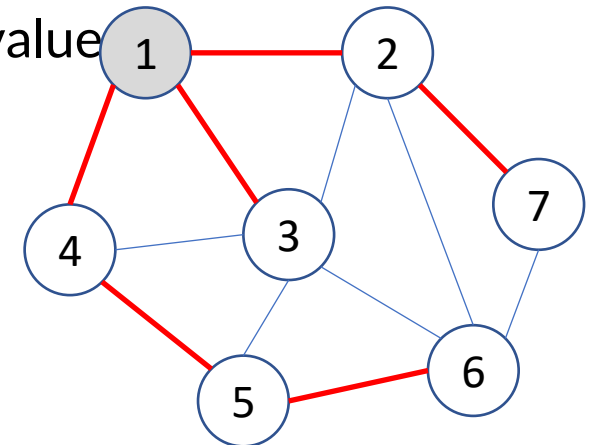
- Objective function

- $f(X, Y, Z) = \sum_{k=1}^K \sum_{(i,j) \in A} X(k, i, j) d(i, j) \rightarrow \min$

EXAMPLES

- **MultiCast Routing Problem**

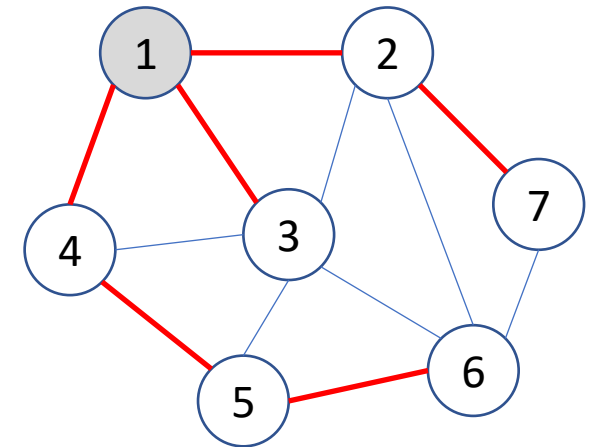
- Given a network $V = \{1, \dots, N\}$ is the set of nodes, $E \subseteq V^2$ is the set of links between nodes. A node $s \in V$ is the source node which will transmit a package to others nodes. A node receiving the package can continue transmit this package to adjacent nodes.
 - $t(i,j)$ and $c(i,j)$ are transmission time and transmission cost when transmitting the package from node i to node j
- Compute the set of links used for broadcasting the package from the source node to all other nodes such that
 - Total transmission time from s to any node cannot exceed a given value
 - Total transmission cost is minimal



- **MultiCast Routing Problem**

- Denote $A(i) = \{j \in V \mid (i,j) \in E\}$, M is a big constant
- Decision variables
 - Binary variable $X(i,j) = 1$ if the package is transmitted from node i to node j , and $X(i,j) = 0$, otherwise, $\forall (i,j) \in E$
 - $Y(i)$: time-point when the package arrives at node i , $\forall i \in V$
- Constraints
 - $\sum_{i \in A(j)} X(i,j) = 1, \forall j \in V \setminus \{s\}$
 - $Y(i) + t(i,j) + M(1-X(i,j)) \geq Y(j), \forall (i,j) \in E$
 - $Y(i) + t(i,j) + M(X(i,j)-1) \leq Y(j), \forall (i,j) \in E$
 - $Y(i) \leq L, \forall j \in V \setminus \{s\}$
 - $Y(s) = 0$
- Objective function to be minimized

$$f(X) = \sum_{(i,j) \in E} c(i,j)X(i,j)$$



- **Facility Location Problem**

- There are M sites $1, 2, \dots, M$ that can be used to open facility for servicing N customers $1, 2, \dots, N$.
 - $f(i)$ is the cost for opening the site i
 - $Q(i)$ is the capacity of site i (maximum amount of good it can serve customers)
 - $c(i,j)$ is the cost for transporting unit of good from site i to customer j
 - $d(j)$ is the total demand (amount of goods) of customer j
- Compute a planning (which site to be opened and amount of good each opened site serves a customer) such that
 - Capacity constraint is satisfied
 - Total cost is minimal

- **Facility Location Problem**

- Decision variables

- $Y(i)$ – binary variable, $Y(i) = 1$ means that the site i is opened, and $Y(i) = 0$, otherwise
 - $X(i,j)$ – amount of good site i serves customer j

- Constraints

- $\sum_{i=1}^M X(i,j) = d(j), \forall j = 1, \dots, N$
 - $\sum_{j=1}^N X(i,j) \leq Q(i)Y(i), \forall i = 1, \dots, M$
 - $0 \leq X(i,j) \leq d(j)Y(i), \forall i = 1, \dots, M, \forall j = 1, \dots, N$

- Objective functions

$$\sum_{i=1}^M f(i)Y(i) + \sum_{i=1}^M \sum_{j=1}^N c(i,j)X(i,j) \rightarrow \min$$

A graphic on the left side of the slide featuring a dark blue background with a large, stylized circular pattern of red dots. The dots are arranged in concentric, slightly irregular rings, creating a textured, halftone-like effect. In the center of this pattern, the word "HUST" is written in a bold, white, sans-serif font.

HUST

THANK YOU !